===

CASE STUDY 1: STUDENT PERFORMANCE ANALYSIS

===

a)

```
students <- data.frame(
Student_ID=c(101,102,103,104,105,106),
Name=c("Anu","Bala","Charan","Divya","Esha","Farhan"),
Department=c("CSE","ECE","CSE","IT","ECE","CSE"),
Gender=c("F","M","M","F","F","M"),
Marks=c(85,72,91,68,88,79)
)
```

b)

```
str(students)
summary(students)
```

c)

```
subset(students, Marks > 80)
```

d)

The data frame is suitable because each row represents one student and each column represents one variable. It supports filtering, grouping, summarization and visualization. Missing values and incorrect data types should be checked before advanced analysis.

===

CASE STUDY 2: HOSPITAL PATIENT RECORDS

===

a)

```
patients <- data.frame(
Patient_ID=c(1,2,3,4),
Age=c(25,40,60,35),
Gender=c("M","F","M","F"),
Disease=c("Flu","Cancer","Diabetes","Fever"),
Admission_Days=c(3,10,8,2),
Treatment_Cost=c(5000,NA,12000,4000)
)
```

b)

```
patients[is.na(patients$Treatment_Cost), ]
```

c)

```
subset(patients, Admission_Days > 7)
```

d)
Missing values should be handled before calculating the average treatment cost. Removing records may reduce the sample size, while replacing missing values with the median is usually appropriate if only a few values are missing randomly.

===
CASE STUDY 3: RETAIL SALES DATA FRAME
===

a)
```
str(sales)
summary(sales)
```

b)
```
sales$Total_Sales <- sales$Quantity_Sold * sales$Unit_Price
```

c)
```
sales[order(-sales$Total_Sales), ]
```

d)
Sorting identifies products with the highest revenue, but profit margin, discounts, returns and inventory turnover should also be considered before deciding the most successful product.

===
CASE STUDY 4: EMPLOYEE SALARY ANALYSIS
===

a)
```
employees[, c("Name","Department","Salary")]
```

b)
Department-wise average salary can be calculated and each employee can be compared with the department average.

c)
```
employees[employees$Salary < mean(employees$Salary, na.rm=TRUE), ]
```

d)
Salary alone is not a complete measure of employee performance. Experience, skills, responsibilities, workload and performance ratings should also be considered.

===
CASE STUDY 5: ENVIRONMENTAL POLLUTION MONITORING

===

a)
summary(pollution)
str(pollution)

b)
pollution[pollution$Air_Quality_Index > 200,]

c)
Possible data-quality problems:
- Missing values
- Incorrect units
- Inconsistent city names
- Duplicate records
- Extreme outliers

d)
The data frame supports preliminary analysis, but long-term environmental policies require more years of data, weather conditions, traffic information and additional pollutants.

===
CASE STUDY 6: ONLINE COURSE ENROLLMENT
===

a)
courses[courses$Enrolled_Students > 1000 &
courses$Completion_Rate < 50,]

b)
High enrollment alone does not indicate course success because learners may not complete the course.

c)
courses$Performance <-
ifelse(courses$Completion_Rate >= 70,"High",
ifelse(courses$Completion_Rate >= 50,"Medium","Low"))

d)
The data frame is useful for identifying problem courses, but ratings, learner feedback and course difficulty should also be analyzed.

===
CASE STUDY 7: BANKING CUSTOMER DATA

==

a)
bank[bank$Credit_Score > 700 &
bank$Balance > 50000,]

b)
These conditions identify financially strong customers but are not sufficient for loan approval.

c)
Additional variables:
- Income
- Existing debt
- Employment status
- Repayment history
- Loan purpose

d)
The data frame is suitable for customer segmentation but not for final loan approval decisions.

==
CASE STUDY 8: TRANSPORTATION ACCIDENT RECORDS
==

a)
accidents[accidents$Severity == "Severe",]

b)
table(accidents$Weather)

c)
table(accidents$Vehicle_Type)

d)
Accident frequency alone should not determine road improvement priorities. Road condition, traffic volume and accident severity should also be considered.

==
CASE STUDY 9: E-COMMERCE CUSTOMER ORDERS
==

a)
orders[orders$Delivery_Days > 5,]

b)
Compare the average customer ratings of delayed and non-delayed orders.

c)
The data frame can identify an association between delivery delay and ratings but cannot prove that delay is the only reason for poor ratings.

d)
Possible confounding factors:
- Product quality
- Customer expectations
- Seasonal demand
- Delivery location
- Return issues

==
CASE STUDY 10: AGRICULTURAL CROP YIELD DATA
==

a)
```
str(crops)
summary(crops)
```

b)
Compare yield values by Crop or Soil_Type using subset() or aggregate().

c)
```
summary(crops$Yield)
boxplot(crops$Yield)
```

d)
The data frame is useful for analysis but is not sufficient to recommend one fertilizer strategy for all farms because yield depends on soil type, rainfall, temperature, irrigation and crop variety.

==
IMPORTANT R FUNCTIONS
==

```
data.frame()    -> Creates a data frame
str()           -> Displays the structure
summary()       -> Summary statistics
head()          -> First six rows
tail()          -> Last six rows
subset()        -> Filters rows
```

```
is.na()             -> Finds missing values
complete.cases()    -> Finds complete rows
order()             -> Sorts rows
mean(...,na.rm=TRUE)-> Mean ignoring NA
[]                  -> Selects rows and columns
```